7 A company has a network that includes a web server.

Data is transmitted across the network using serial half-duplex data transmission.

(a) Draw and annotate a diagram to show how the data is transmitted using serial half-duplex data transmission.

[4]

(b) Give **two** benefits of the company using serial half-duplex data transmission for this purpose.

1 .....

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2 .....

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[2]

(c) Give **one** drawback of the company using serial half-duplex data transmission for this purpose.

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..... [1]

(d) The company uses a proxy server to help protect the web server and the network from cyber security threats.

(i) Give **three** cyber security threats that the proxy server can help protect against.

1 .....

2 .....

3 .....

[3]

(ii) Identify **two** functions of the proxy server that can be used to help protect the web server and the network.

1 .....

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2 .....

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[2]

(e) Customers access the company's web pages that are stored on the web server.

Describe how the web pages are located, retrieved and displayed on a customer's computer.

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..... [6]